# Aligned Mathematical Structures

Source-backed grouped formula sample

## 1 Expressions

**Expression 1.** The following expression is taken from a source-backed formula corpus.

$$\begin{bmatrix} V_{\ell,YY} & V_{\ell,YX} \\ V_{\ell,XY} & V_{\ell,XX} \end{bmatrix} = \begin{bmatrix} (Q_\ell/Q_{\ell,XX})^{-1} & -Q_{\ell,YY}^{-1}Q_{\ell,YX}(Q_\ell/Q_{\ell,YY})^{-1} \\ -(Q_\ell/Q_{\ell,YY})^{-1}Q_{\ell,XY}Q_{\ell,YY}^{-1} & (Q_\ell/Q_{\ell,YY})^{-1} \end{bmatrix},$$

**Expression 2.** The following expression is taken from a source-backed formula corpus.

$$\left\{ \begin{array}{rcl} \mathrm{d}X_s & = & u_s \mathrm{d}W_s, \\ -\mathrm{d}Y_s & = & \Big[ \langle Q_s X_s, X_s \rangle + \langle R_s u_s, u_s \rangle + \kappa Z_s \Big] \mathrm{d}s - Z_s \mathrm{d}W(s), \\ X_0 & = & x, Y_T = \langle G X_T, X_T \rangle , \end{array} \right.$$

**Expression 3.** The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} &\int_{\mathbb{T}^d \times [0,T)} \bar{f} \left( \frac{\partial \phi}{\partial t} + u \cdot \nabla \phi \right) \, dxdt \\ &= \lim_{i \to \infty} \int_{\mathbb{T}^d \times [0,T)} f^{\nu_i} \left( \frac{\partial \phi}{\partial t} + u \cdot \nabla \phi + \nu_i \Delta \phi \right) \, dxdt \\ &= - \int_{\mathbb{T}^d} f_0 \phi_0 \, dx, \end{aligned}$$

**Expression 4.** The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} |\frac{f(p)}{\xi(p)} - \frac{f(q)}{\xi(q)}| &= |f(p)(\frac{1}{\xi(p)} - \frac{1}{\xi(q)}) + \frac{f(p)-f(q)}{\xi(q)}| \\ &\leq 2\xi(p)(\frac{1}{\xi(p)} - \frac{1}{\xi(q)}) + \frac{d(p,q)}{\xi(q)} \\ &= \frac{1}{\xi(q)}[2(\xi(q)-\xi(p)) + d(p,q)] \\ &\leq 3\frac{d(p,q)}{\xi(q)} = 3\rho(p,q). \end{aligned}$$

**Expression 5.** The following expression is taken from a source-backed formula corpus.

$$\begin{multline*}\gamma_*\bigl(\gamma^*(x,y)\otimes {\alpha^*}_\rho(z)\bigr)=\gamma_*(x,y)((z-x)e)\\ (x,y)\otimes\rho_{*\rho}(z)-(x-3,y-1)\otimes\rho_{*\rho}(z-1)\\ \sum_{0\le h\le z}(x-h,y)-\sum_{0\le k\le -z-2}(1+x+k,y)-\sum_{0\le j\le z-1}(x-3-j,y-1)+\sum_{0\le l\le -z-1}(x+l-2,y-1)\end{multline*}$$